FWO

WRITING A FINAL DMP

Ghent University Data Stewards

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BKT

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https://osf.io/avjmu

OUTLINE

- FWO policy
- Data lifecycle and DMP
- Content of final DMP
- Register in Biblio
- Wrap-up
- Q&A

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Opening new horizons

- RDM = key part of FWO policy for all support channels
  → "Sound data management is required for all research supported by the FWO."
- Specific expectations:
  1. preserve data for a minimum of 5 years after end of research
  2. submit a Data Management Plan
  3. encourages researchers to ensure that their data management complies as much as possible with the 'FAIR' principles
  4. advises researchers to apply the principle 'as open as possible, as closed as necessary'
- Reporting
  → the FWO obliges to register all relevant output in the FRIS database via the UGent institutional systems Biblio and GISMO.

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https://www.fwo.be/en/the-fwo/research-policy/data-management-plan/
https://www.fwo.be/en/the-fwo/research-policy/reporting/

FAIR DATA AND SOFTWARE

Findable → How easily can humans (and machines) find my data? How good are my metadata?

Accessible → Once found, how easy is it to access my data? Are the metadata accessible, even if the data aren't?

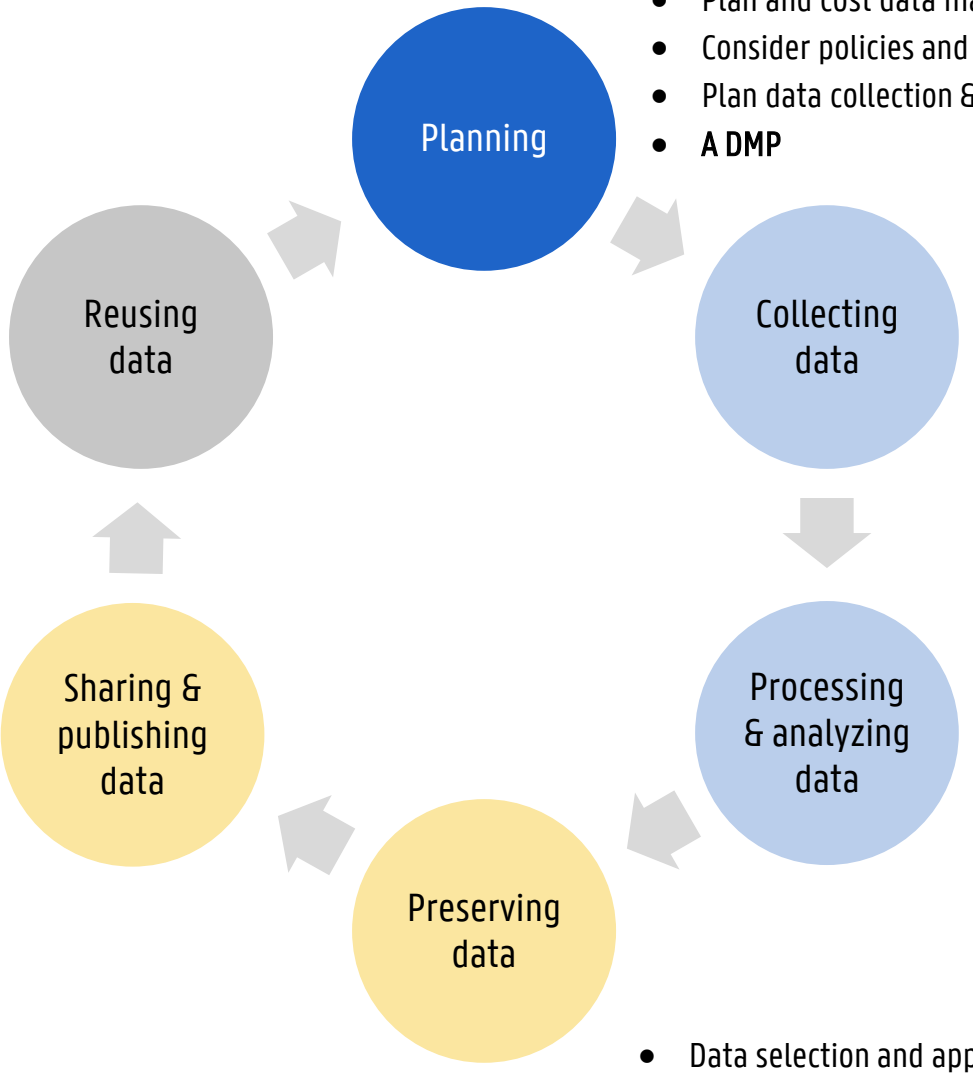
Interoperable → How well does my data integrate with other data? Is a standard vocabulary used?

Reusable → How difficult is it to reuse my data? Are the (meta)data well described? Is there a usage license?

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https://www.ugent.be/en/research/datamanagement/after-research/fair-data.htm

RESEARCH DATA LIFECYCLE



- Design research
- Plan and cost data management
- Consider policies and regulations
- Plan data collection & processing, security measures
- **A DMP**

- Data collection: experiments, observations, surveys, modelling, compiling, 3rd-paty data
- Metadata generation
- Documenting data acquisition
- Data quality assurance.

- Entering, digitizing, transcribing, deriving, anonymizing
- Data transformation, harmonization and integration
- Quality checking, validation, cleaning
- Analysis and interpretation
- Describing and documenting
- Storing, organizing, version control

- Data selection and appraisal
- Migrating to open, sustainable formats
- Archiving data and documentation in appropriate location

- Choose discipline specific or generic repository
- Provide documentation and discovery metadata
- Choose license and access control
- Publish y a data paper

- Find data
- Check access and use conditions
- Cite data

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https://www.youtube.com/watch?v=0L_Vd9dd-AQ

FWO DMP REQUIREMENT

1. **Proposal stage**
   → 'DMP-light': 5 RDM questions in FWO application form
   → not an evaluation criterion (but expert panels/referees may comment)
2. **Post-award**
   → submit full DMP to host institute no later than 6 months after official start date
   → use template
   → keep it up to date
3. **Upon completion**
   → append final version of DMP to final project report
   → DMP is part of final project evaluation
   → new support from the FWO is no longer possible until this obligation has been complied with

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https://www.fwo.be/en/the-fwo/research-policy/data-management-plan/
https://www.fwo.be/en/the-fwo/research-policy/reporting/

DMP'S - GOOD PRACTICE FOR ANY PROJECT

Consider it a tool to help you

- increase awareness of RDM
- ensure data are of high quality & remain usable
- comply with legal, ethical & other requirements
- identify any support & resources required
- clarify expectations & responsibilities

- DMPs take some time & effort upfront, but can save you headaches later on!

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CONTENT OF FINAL DMP

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CONTENT OF A DMP

Sections:

- Research data summary
- Documentation and metadata
- Data storage & back-up during the project
- Data preservation after the end of the project
- Data sharing and reuse
- Responsibilities

END OF PROJECT

Revise and update

Document the output of the project: managed research data

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Training slides: https://osf.io/pfyj5

More info: https://www.ugent.be/en/research/datamanagement/before-research/datamanagementplan.htm

DMP: FIRST VERSION VS. LAST VERSION

Data Management Plan
First submission

- Basic information about your data.
- Data storage and back up vs. long term preservation.
- Initial idea about metadata standards and dedicated archive / repository.
- Identification of legal and/or ethical issues.
- Establishing responsibilities.
- First scope of reusability.

Data Management Plan
Final submission

- Precise description of all data types.
- Clear long-term preservation strategy: selected data archive(s) and/or repository(ies).
- FAIR data and open data ("As open as possible, as closed as necessary").
- Definite selection of reusable outputs.

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Adapted from: Jasmin Böhmer. (2019, September). UBC RDM Workshop 1 - Data Management Planning (Version V2). Zenodo. http://doi.org/10.5281/zenodo.3469953

RESEARCH DATA SUMMARY

- Verify that the list of research data or other research outputs is complete
  - Make sure all data types are accurately described such that other researchers can understand the data overview

- Capture changes in policies or strategies regarding ethical or legal issues: e.g. new innovation potential, decision to file for a patent, …

EXAMPLE DATA TABLE

Dataset name	Description	New or reused	Data type	Data format	Volume
WP1 Nutrient bio-availability	Comparison of nutrient bio-availability after different thermochemical treatments	New	Experimental, quantitative	xlsx, csv	<5MB
WP1 Data from statistical analysis	Results from statistical analysis to determine the influence of the dependent variables (chemical treatment/reactors, temperature, retention time) in pollutant removal and nutrient bio-availability.	New	Analysis script & results, quantitative	R, csv, png	<20MB
WP2 Consumers attitudes - questionnaire responses	Data on attitudes towards consumption of food that has been grown using fertilisers based on recycled human excreta. Online questionnaires will be performed using Qualtrics and will collect: ● demographic data from participants: place of residence, age, gender, education, occupation, annual income. ● respondent's opinions or views about issues in relation to the research topic: environmental awareness questions, attitudes or acceptance of the proposed product, etc.	New	Observational, quantitative & qualitative	csv	<5MB
WP2 Consumer attitudes - analysis	Statistical analysis of survey responses	New	Analysis results, quantitative	sav	<10MB

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DOCUMENTATION AND METADATA

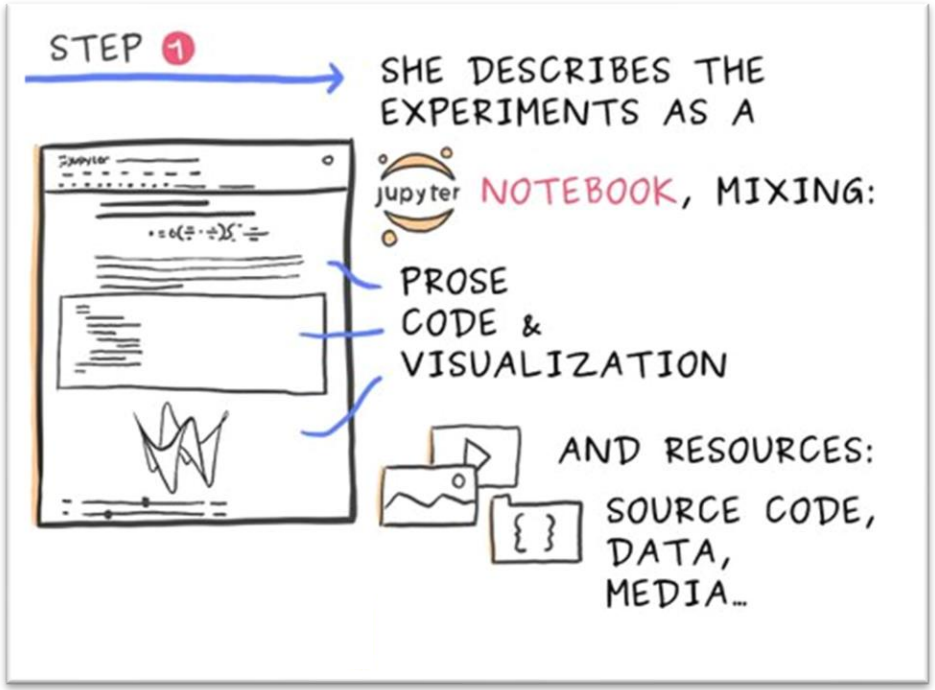
- Verify that the documentation methods are accurately described.
- Provide in the DMP the location of the data documentation

```
project_name/
├── README.md            # overview of the project
├── data/                # data files used in the project
│   ├── README.md        # describes where data came from
│   └── sub-folder/      # may contain subdirectories
├── processed_data/      # intermediate files from the analysis
├── manuscript/          # manuscript describing the results
├── results/             # results of the analysis (data, tables, figures)
├── src/                 # contains all code in the project
│   ├── LICENSE          # license for your code
│   ├── requirements.txt # software requirements and dependencies
│   └── ...
└── doc/                 # documentation for your project
    ├── index.rst
    └── ...
```

Readme files



Paper and electronic lab notebooks



Jupyter notebooks
https://opendreamkit.org/2017/11/02/use-case-publishing-reproducible-notebooks/

Codebook summary table

Label	Variable	Class	# unique values	Missing	Description
Sepal length in cm	**Sepal.Length**	numeric	35	0.00 %	Measured using a line gauge produced by Acme factories.
	Sepal.Width	numeric	23	0.00 %	
	Petal.Length	numeric	43	0.00 %	
	Petal.Width	numeric	22	0.00 %	
	Species	factor	3	0.00 %	Two of the three species were collected in the Gaspé Peninsula all from the same pasture, and picked on the same day and measured at the same time by the same person with the same apparatus

Codebooks

DATA PRESERVATION AFTER THE END OF THE PROJECT

Clearly describe the long term preservation strategy:

- What data must be kept and for how long?
- What are the selected data archive(s) or repositories? How can the data be found?
- How are the data organized?
- Is data available in sustainable formats?
- Is there a 'data package' of data, documentation and metadata?
- Are the proper access rights in place?
- Is it clear what reuse is permitted?



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https://www.ugent.be/en/research/datamanagement/after-research/preservation.htm

DATA SHARING AND REUSE

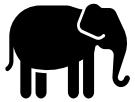
“Open data are publicly accessible and reusable by anyone. However, there may be valid reasons to restrict the access to data (i.e. privacy, intellectual property rights, ethical aspects, dual use, etc.). FWO advises researchers to apply the principle 'as open as possible, as closed as necessary'.” – FWO



Personal data



Confidential data



Sensitive data



Third-party data



Copyright



Valorization

DATA SHARING AND REUSE

- Define the final selection of reusable outputs and mention the reuse conditions: e.g. applicable license, conditions for data transfer contracts or Data Access requirements
- Document any reasons for restricting access and/or reuse of the data.

Dataset is deposited in an online repository

- Provide Digital Object Identifier (DOI) or other persistent identifier or accession number (if available) of the dataset
- Provide the repository of the dataset
- Describe the access level and license of the dataset
- Describe when the dataset will be available

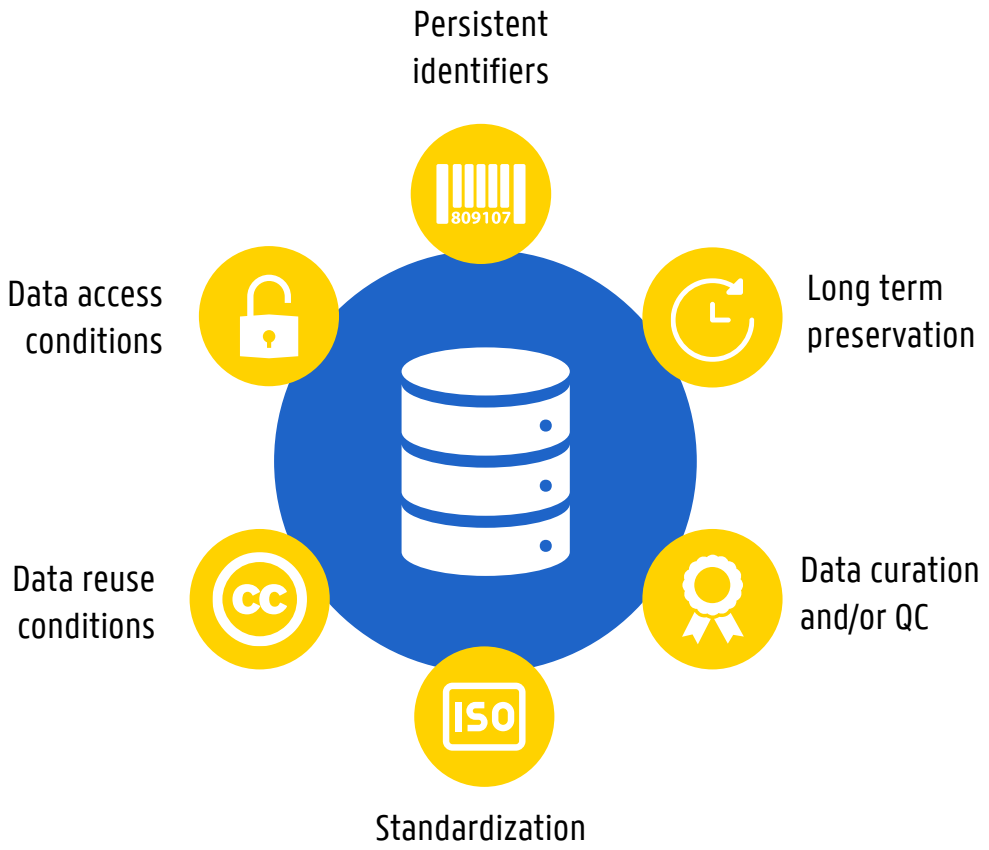
Dataset is NOT deposited in an online repository

- Provide storage location on UGent storage solutions
- Describe implemented technical and organizational measures for data security: access control, read/write permissions, encryption keys, pseudonymization keys, etc.

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WHAT IS A DATA REPOSITORY?

A database infrastructure that collects, manages and stores data for preservation and sharing



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More information: https://osf.io/pb94m

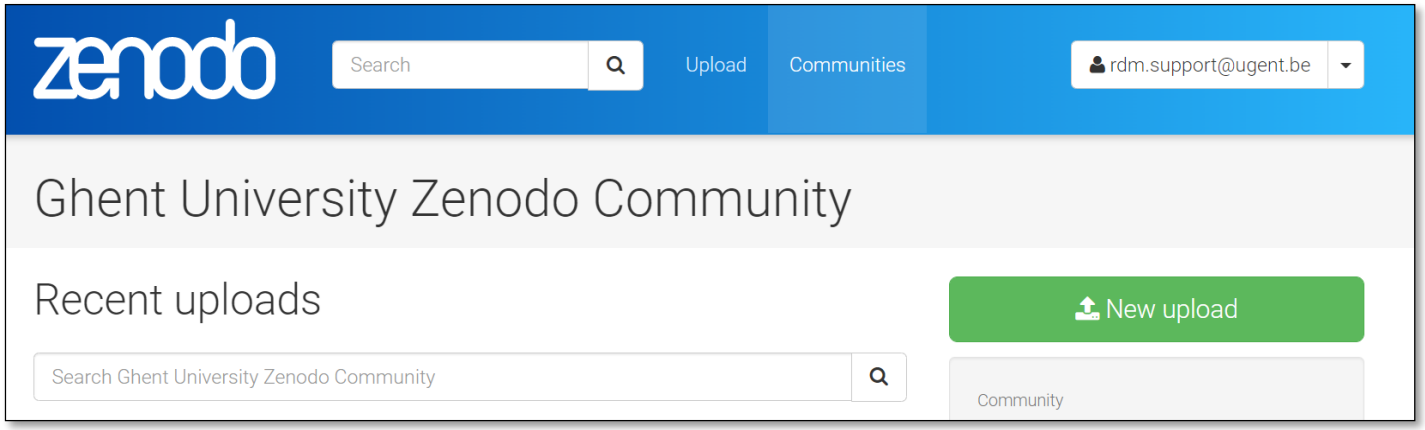
ZENODO.ORG

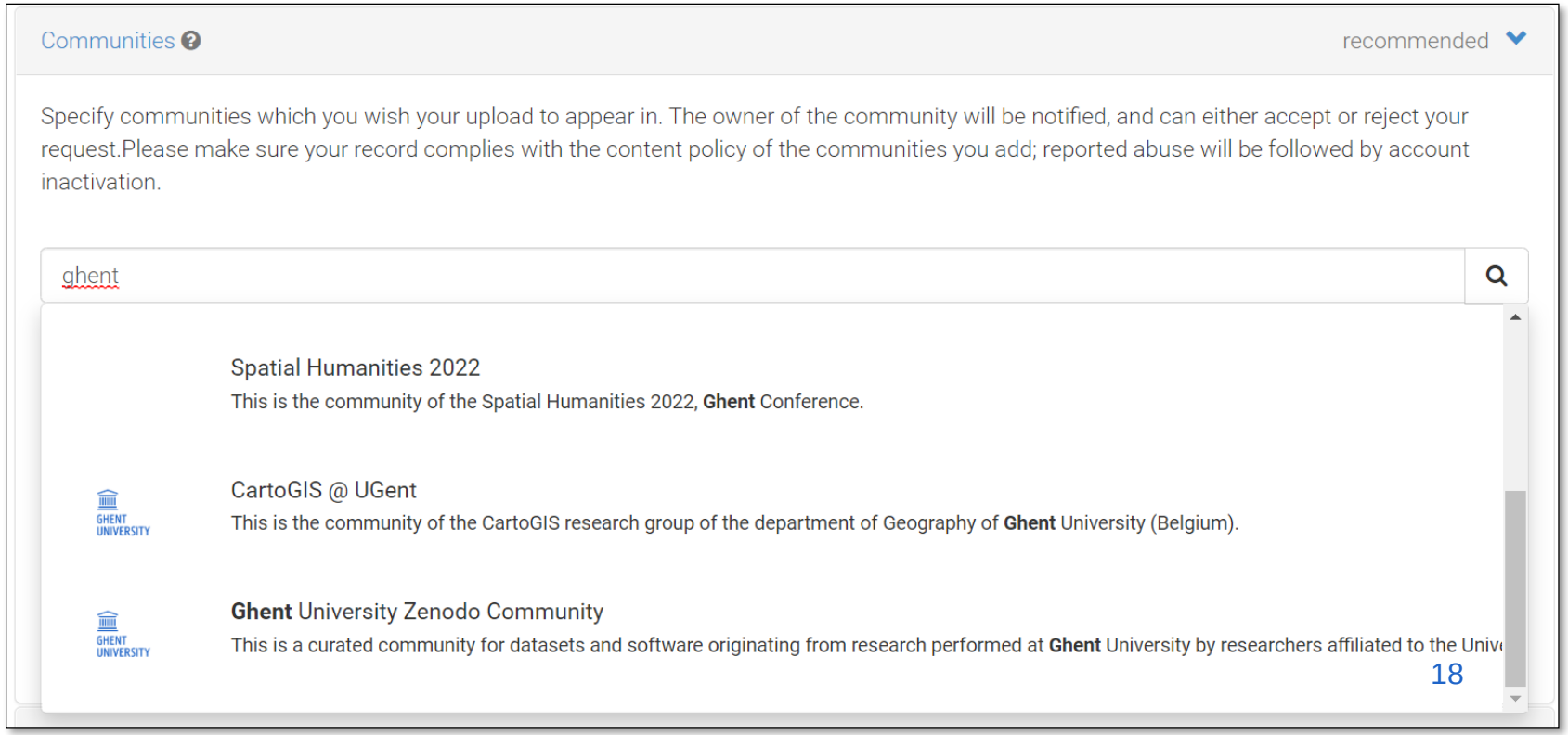
Generic repository for data

- Accepts research data and software
- Provides DOI

Community:

- Curation by Data Curators
- Avoid spam filters and size limitation





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RESPONSIBILITIES

- Indicate the roles and responsibilities for research data management after the research project and when you/researchers leave the project



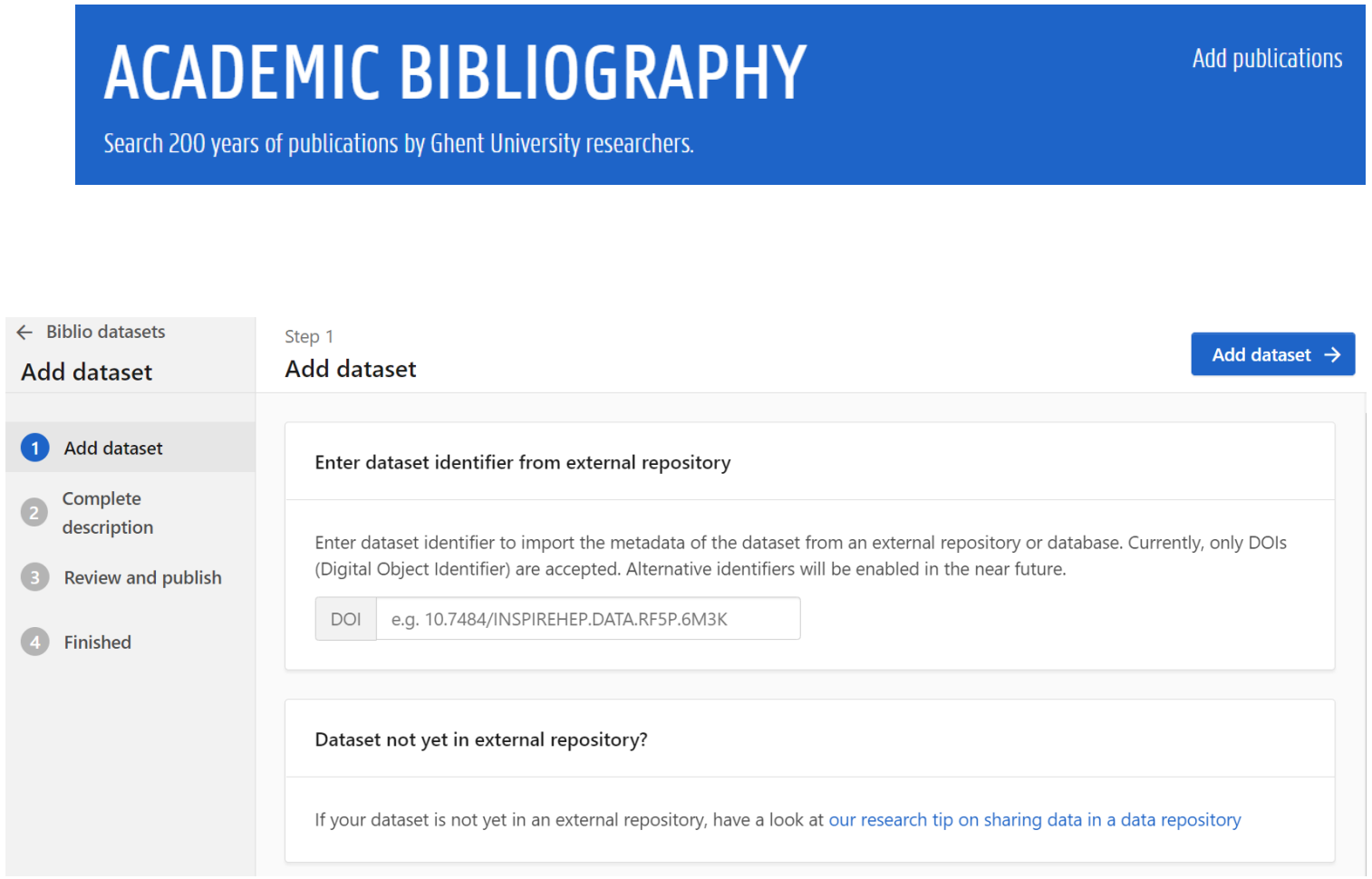
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REGISTER RELEVANT OUTPUT BIBLIO

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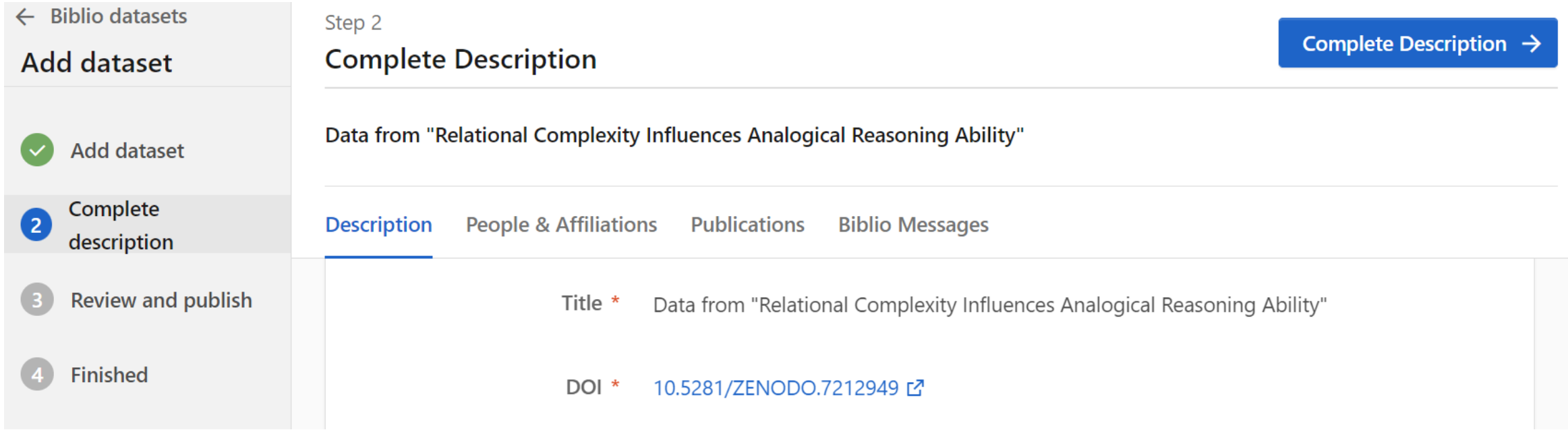
BIBLIO: DATASETS AND SOFTWARE

- Register published research datasets in Biblio using DOI
- Automatically harvests several metadata fields
- Mandatory with new UGent policy for datasets underlying publications
- Enhance findability of your research outputs



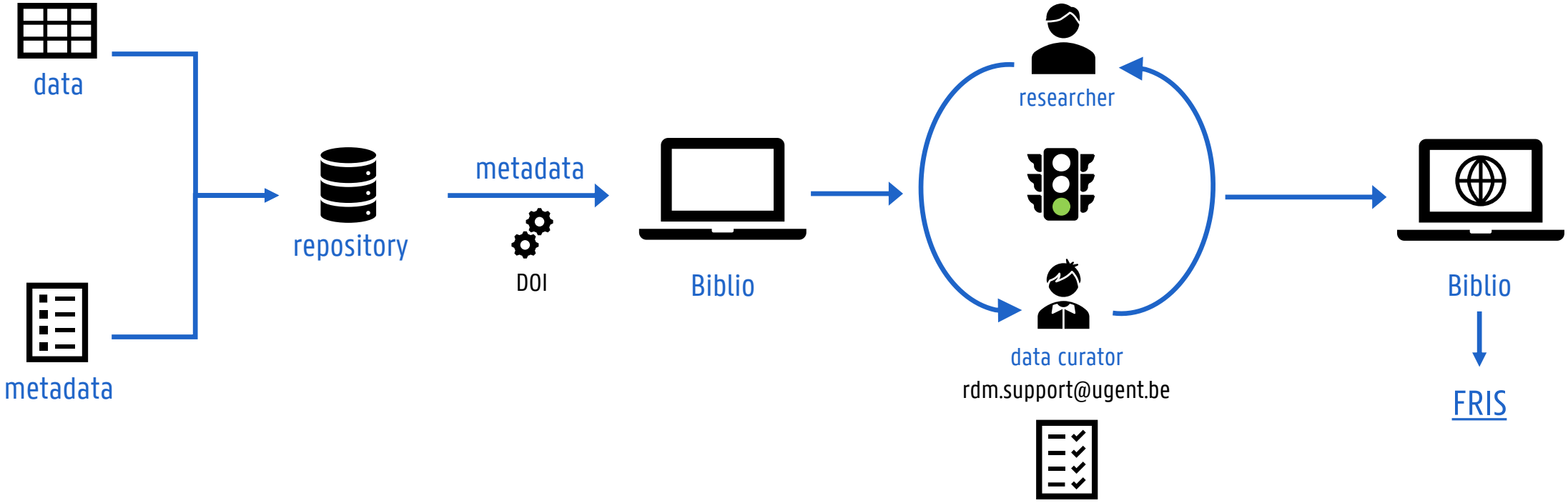
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BIBLIO: DATASETS AND SOFTWARE



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BIBLIO REGISTRATION WORKFLOW FOR DATASETS AND SOFTWARE



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https://www.fwo.be/en/the-fwo/research-policy/research-results/
https://onderzoektips.ugent.be/en/tips/00002054/

WRAP-UP

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TAKE-AWAY MESSAGES

- Deposit data or other research output in repository when possible
  - or explain why it is not possible
- Revise and update DMP
- Append final version of DMP to final project report
- Register in Biblio

Research tip: https://onderzoektips.ugent.be/en/tips/00002153/

UGent Data Stewards

RESEARCH DEPARTMENT - UNIVERSITY LIBRARY


rdm.support@ugent.be

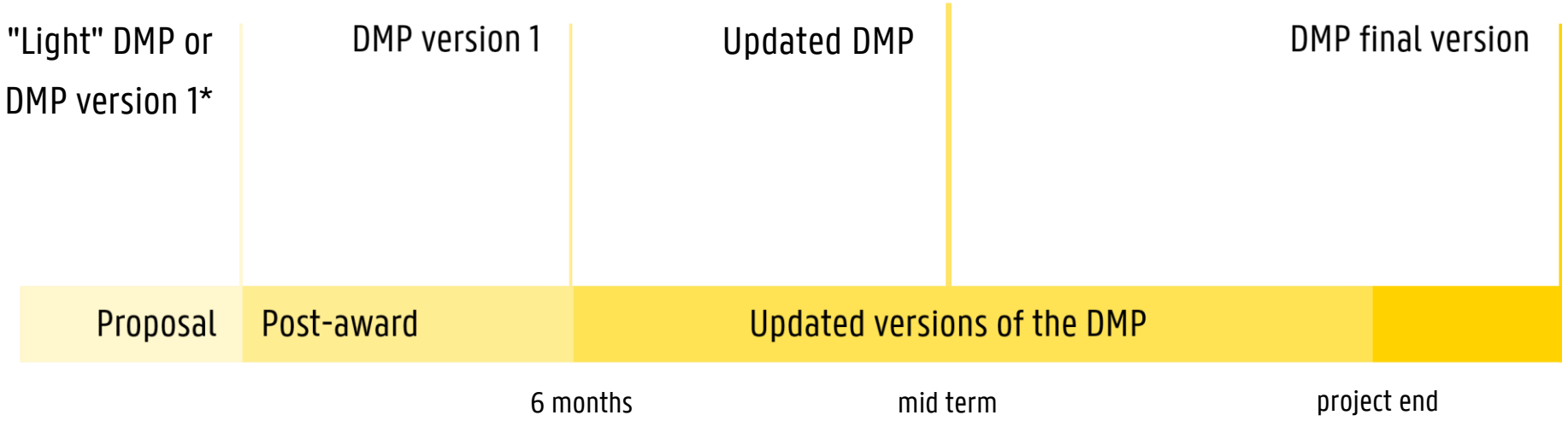

@UGentRDM


ugent.be/en/research/datamanagement

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ESTABLISH + REGULARLY UPDATE A **DATA MANAGEMENT PLAN** ('DMP') – HORIZON EUROPE

- DMP as a deliverable of the project
- DMP is evaluated by reviewers
- Data management costs can be budgeted
- Project coordinator compiles data overview and data management plans from project contributors, discussion on approach for making and executing the DMP during project kick-off meeting is recommended
- Final DMP describes the implementation i.e. how datasets are shared and made available for re-use



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* in case of public emergency

ESTABLISH + REGULARLY UPDATE A **DATA MANAGEMENT PLAN** ('DMP')

Data Management Plan
Proposal

Data Management Plan
First submission

Data Management Plan
Final submission

1 page on how the data/research outputs will be managed in line with the FAIR principles

- Types of data/research outputs
- Findability
- Accessibility
- Interoperability
- Reusability
- Curation and storage/preservation costs

Full DMP in case of public emergency or other requirements due to nature of project

Plan on how to implement FAIR data principles

- Types of data/research outputs
- Potential data repositories
- Potential standards for data and metadata (formats and vocabularies)
- Documentation strategy
- Identification of legal and/or ethical issues and associated data protection/security measures
- Established responsibilities

Description of FAIR data implementation

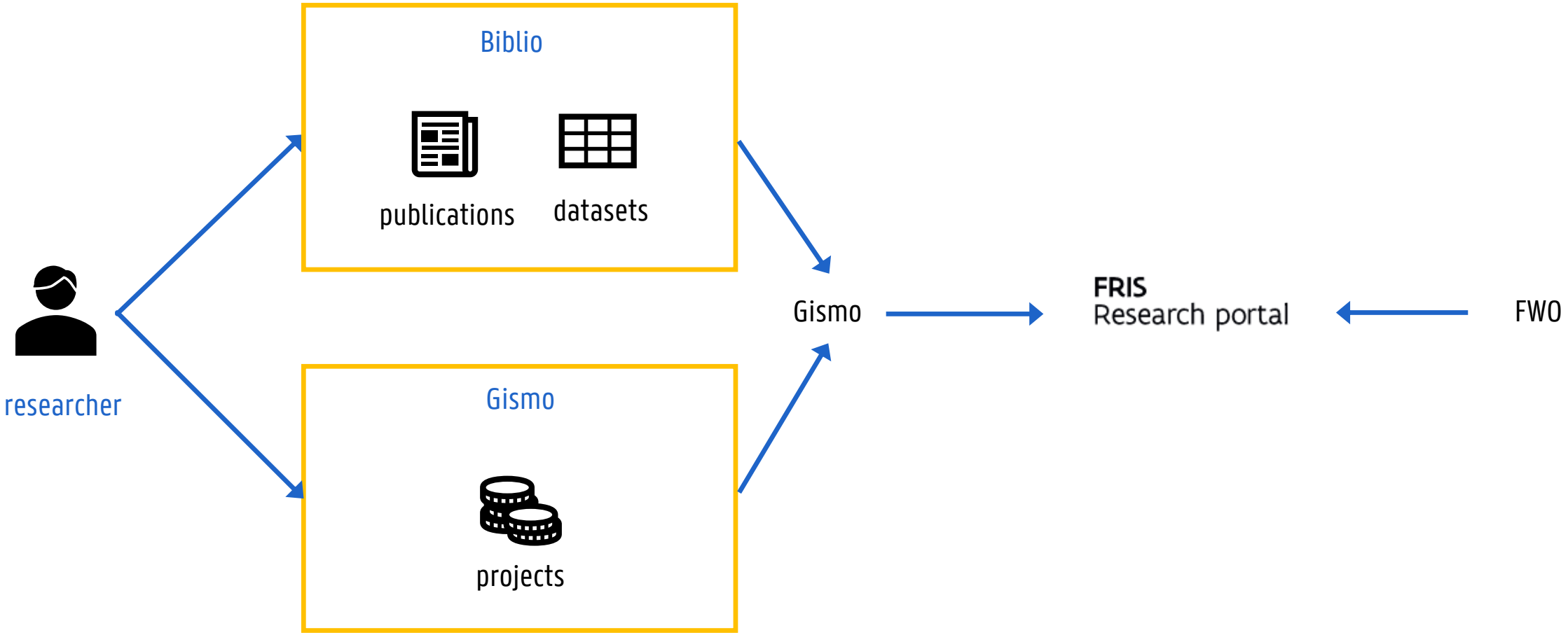
- Precise description of all datasets with identifiers and metadata, including keywords, licenses, access conditions
- Data repositories used to share/preserve the data
- Details on documentation content and format
- Description of applied standards (formats and vocabularies)

When significant changes arise, including, but not limited to:

- generation of new data
- changes in data access provisions or curation policies
- attainment of tasks (e.g. datasets deposited in a repository, etc.)
- changes in relevant practices (e.g. new innovation potential, decision to file for a patent)
- changes in consortium composition

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REPORTING ON RESEARCH OUTPUT



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